

ADCA – Semester 1 Detailed Syllabus

Subject 1: Computer Fundamentals & Operating System

Unit 1: Introduction to Computers

- Definition of Computer
- Characteristics of Computer
- Applications of Computer
- Advantages and Limitations
- History of Computers
- Generations of Computers (1st to 5th Generation)
- Types of Computers
 - Analog
 - Digital
 - Hybrid
 - Micro Computer
 - Mini Computer
 - Mainframe Computer
 - Super Computer

Unit 2: Computer Hardware

- Input Devices
 - Keyboard
 - Mouse
 - Scanner
 - Barcode Reader
 - MICR
 - OCR
 - OMR
 - Webcam
- Output Devices

- Monitor
- Printer
- Plotter
- Speaker
- Storage Devices
 - Hard Disk
 - SSD
 - CD/DVD
 - Pen Drive
 - Memory Card

Unit 3: Computer Memory

- Primary Memory
 - RAM
 - ROM
- Secondary Memory
- Cache Memory
- Virtual Memory
- Memory Units
 - Bit
 - Byte
 - KB
 - MB
 - GB
 - TB

Unit 4: Software Concepts

- System Software
- Application Software
- Utility Software
- Programming Software
- Firmware

Unit 5: Operating System

- Introduction to Operating System
- Functions of OS
- Types of OS
- Single User OS
- Multi User OS
- Batch Processing
- Time Sharing
- Real Time OS

Unit 6: Windows Operating System

- Desktop
- Taskbar
- Start Menu
- File Explorer
- Control Panel
- Settings
- User Accounts
- Device Manager
- Recycle Bin
- Windows Shortcuts

Practical

- Folder Creation
- File Management
- Control Panel Operations
- Windows Settings
- Shortcut Keys Practice
- Disk Management

Subject 2: Computer Architecture & Internet Applications

Unit 1: Number System

- Decimal Number System
- Binary Number System
- Octal Number System
- Hexadecimal Number System
- Number Conversion

Unit 2: Boolean Algebra

- Boolean Operations
- AND Gate
- OR Gate
- NOT Gate
- NAND Gate
- NOR Gate
- XOR Gate
- Truth Tables

Unit 3: Computer Architecture

- CPU
- ALU
- Control Unit
- Registers
- Bus Structure
- Motherboard
- SMPS
- Expansion Slots

Unit 4: Internet Fundamentals

- What is Internet
- History of Internet
- WWW
- Web Browser
- Search Engine

- URL
- Hyperlink

Unit 5: Communication

- Email
- Email Account Creation
- CC & BCC
- Attachments
- Cloud Storage
- Video Conferencing

Unit 6: Internet Security

- Virus
- Malware
- Trojan
- Worm
- Spyware
- Antivirus
- Cyber Safety

Practical

- Gmail Creation
- Sending Email
- Google Drive Usage
- File Upload/Download
- Internet Browsing
- Online Form Filling

Subject 3: MS Word (Word Processing Application)**Unit 1: Introduction to MS Word**

- Word Interface
- Ribbon

- Tabs
- Groups
- Quick Access Toolbar

Unit 2: Document Management

- Create Document
- Open Document
- Save Document
- Print Document

Unit 3: Text Formatting

- Font Formatting
- Paragraph Formatting
- Alignment
- Line Spacing
- Bullets & Numbering
- Borders & Shading

Unit 4: Page Formatting

- Margins
- Orientation
- Page Size
- Header & Footer
- Watermark
- Page Border

Unit 5: Insert Features

- Table
- Picture
- Shapes
- WordArt
- SmartArt
- Symbols
- Hyperlinks

Unit 6: Advanced Features

- Mail Merge
- Find & Replace
- Track Changes
- Comments
- Templates

Practical

- Resume Design
- Letter Writing
- Certificate Design
- Application Form Design
- Mail Merge Project

Subject 4: MS Excel (Electronic Spreadsheet Application)

Unit 1: Introduction to Excel

- Workbook
- Worksheet
- Rows & Columns
- Cell Address

Unit 2: Data Entry

- Editing Data
- Auto Fill
- Series
- Data Validation

Unit 3: Formatting

- Cell Formatting
- Conditional Formatting
- Borders
- Alignment

Unit 4: Formula & Functions

- Formula Basics
- Relative Reference
- Absolute Reference

Important Functions

- SUM()
- AVERAGE()
- MAX()
- MIN()
- COUNT()
- IF()
- ROUND()
- TODAY()
- DATE()

Unit 5: Data Analysis

- Sorting
- Filtering
- Subtotal
- Pivot Table (Basic)

Unit 6: Charts

- Column Chart
- Bar Chart
- Pie Chart
- Line Chart

Practical

- Marksheet Preparation
- Salary Sheet
- Attendance Sheet
- GST Calculation
- Sales Report

- Chart Creation

Internal Assignment / Project

Students are generally required to prepare:

- Computer Fundamentals Assignment
- Internet Assignment
- MS Word Project
- MS Excel Project
- Practical File
- Viva Voce Preparation

Semester-1 Outcome

After completing Semester-1, a student should be able to:

- Operate Windows efficiently
- Manage files and folders
- Use Internet and Email professionally
- Create professional documents in MS Word
- Prepare reports and calculations in Excel
- Understand computer architecture and number systems
- Apply basic cybersecurity practices